Are Weyl Gravity's Ghosts Real? Building Model Universes to Find Out

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Einstein's theory says that gravity is the shape of spacetime. Matter bends spacetime; the resulting geometry guides planets, light and clocks. This picture explains gravitational redshift, the bending of light, gravitational waves, black holes and the expansion of the universe.

Weyl gravity begins with a more symmetrical set of equations. Unlike Einstein's equations, they contain four derivatives of the geometry rather than two. That feature makes the theory interesting as a possible route toward quantum gravity, but it also creates additional gravitational solutions. In ordinary treatments, some of those solutions carry the wrong sign. Physicists call them *ghosts*.

A ghost in an equation is not automatically a ghost in nature. It may be a redundant description, a wave forbidden by a global constraint, a solution excluded by a physical boundary condition, or a genuine instability. Our project is building a sequence of exactly controlled model universes to find out which answer applies—and whether the properties discovered separately can ultimately coexist in one theory.

The rule is simple:

> A pattern in an equation is not yet a physical object. It must survive the constraints, propagate causally, affect a possible observation, and remain consistent when interactions and quantum physics are included.

This article explains what those tests mean, what the project has learned, and what remains unknown. It is a progress report about a research programme, not a claim that Weyl gravity already describes our universe.

The ghost test

An equation can contain more solutions than nature contains physical states. Before calling an extra solution a particle—or declaring it fatal—we ask six questions.

1. **Is it only gauge?** Different mathematical descriptions may represent the same physical situation.
2. **Does it satisfy the constraints?** A solution of the easiest linear equation may fail the full set of gravitational balance laws.
3. **Does it propagate causally?** A source in the future must not change a detector in the past.
4. **Can it affect an observation?** A physical direction should couple to a clock, detector, flux or another relational measurement.

5. **Does it survive interactions?** A small wave must fit into the nonlinear theory without violating a constraint or growing uncontrollably.
6. **Can it support a consistent quantum state?** Quantum probabilities must be positive after all redundant descriptions have been removed, and the quantum gauge identities must remain valid.

These are separate tests. Passing one does not silently grant the others. A classical wave is not yet a particle. A nonzero classical sign is not yet a quantum probability. A regular black-hole solution is not yet a stable black hole.

Gauge: when two descriptions are one reality

Gauge is the physicists' word for redundancy in a description. The same place on Earth can be described by latitude and longitude, a street map or a rotated grid. The numbers change, but the place does not.

Gravity has the same feature. Relabeling the coordinate grid changes the components used to describe spacetime without necessarily changing spacetime itself. Weyl gravity has an additional freedom: one may change the local scale—the choice of ruler—while preserving the light cones.

A gauge transformation is therefore not a new event, force or wave. Counting it as physical would count the same universe more than once. But a transformation that resembles gauge can instead be a physical symmetry. The difference is a measurable generator or charge, such as energy or angular momentum. The calculation must decide which case applies.

This matters especially for time. Saying that a shift of the *overall time label* is gauge in one closed, zero-total-charge model does not mean that clocks stop or change is unreal. Internal clocks can keep changing, and one can ask what another field does when a clock reads a particular value. In a larger sector where the time shift carries energy, the same transformation is a physical symmetry rather than gauge.

Causality: tomorrow must not alter yesterday

A causal theory does not allow a future event to change the past. If a source is switched on tomorrow, a detector must not respond today. A disturbance may influence only events inside its future light cone.

The physical response used for this test is the *retarded* response: it begins at the source and travels toward the future. Physicists also construct an *advanced* mathematical partner. It is useful for checking the equations and building comparison rules, but it does not authorize backward-in-time signalling. The physical source-to-detector rule remains retarded.

For gravity, it is not enough to test one convenient wave equation. The constraints and gauge relations must travel consistently with the wave. One of this project's main technical achievements is to test causality at that full-system level.

Several laboratories, one eventual theory

The final ambition is not to keep a collection of unrelated miniature universes. It is to determine whether one coherent theory can describe causal spacetime, light, clocks, gravitational waves, interactions, black holes and quantum states at the same time.

Trying to test all of that in one calculation would hide the cause of every success or failure. Experimental science therefore uses wind tunnels, vacuum chambers and test tracks before asking one aircraft to fly. We use mathematical laboratories in the same way. Each laboratory removes selected complications and makes one physical requirement exact enough to challenge by calculation.

Most of these laboratories place the same core Weyl equations on different spacetime backgrounds. A *background* is the controlled starting geometry, not a different law of nature. Other laboratories deliberately add light, a clock or a possible quantum repair. Those additions are candidate extensions of the theory and must be declared explicitly.

Model laboratory	Requirement isolated for the eventual theory
Closed spherical universe	Can the complete gravity-and-gauge system propagate causally, and what remains after redundant descriptions and global charges are removed? Space is finite but has no edge, like the surface of Earth one dimension higher. Its history is called the conformal cylinder.
Berger clock universe	Can matter provide clocks, can light carry a causal signal between them, and can redshift and the first interactions be defined without depending on coordinates? Its spatial sphere is deliberately deformed in a controlled way.
Compact circle-by-sphere universe	Which ordinary and additional gravitational waves survive global balance, nonlinear continuation and resonance when electromagnetism is present?

Model laboratory	Requirement isolated for the eventual theory
Nariai curved universe	Is the causal construction a reusable geometric mechanism rather than an accident of the closed spherical universe?
Schwarzschild exterior	Does the same theory contain black holes, and do horizons and distant radiation conditions retain or eliminate the additional fourth-order waves?
Euclidean spherical calculation	Do the quantum symmetry identities survive a one-loop test, and if not, what change of theory would be required to repair them? This is a quantum calculation tool, not a real-time universe with observers.

Each experiment contributes a requirement, not a detachable piece that can simply be pasted into a final universe. The unification task is to place the results beneath one common set of equations and ask whether the relevant constructions agree where their domains overlap:

> **Targeted model universes produce certified physical requirements. Those requirements must then fit into one compatible theory—or produce a precise no-go.**

For example, the gravitational wave called Einstein-like in the compact laboratory must be identified with the corresponding wave near a black hole; the causal rule used for the clock signal must follow from the same gauge system; and any quantum repair must preserve rather than silently replace the classical observables already constructed. Interactions must then show whether these sectors remain compatible or force one another to change.

Results therefore do not automatically transfer between laboratories. A wave constrained in the compact circle-by-sphere universe has not thereby been removed from a Schwarzschild exterior, and a boundary condition that works in a Euclidean calculation is not automatically a causal real-time rule. Every public claim is labeled by its theory, background, charge sector, boundary conditions and stage of completion.

This separation is not a weakness to be hidden. It is how the project learns what the final theory must integrate. If all the requirements can be made compatible, the laboratories become different solutions and experiments inside one theory. If they cannot, the conflict identifies exactly where the proposed universe fails. A unified no-go result would also be a successful scientific outcome.

Three things we have learned

1. Causality belongs to the whole gauge system

The project has constructed exact classical retarded and advanced responses for the complete linked gravitational system on the closed spherical background. Separate constructions pass the same causal test in the Berger clock universe and in the Nariai curved universe. The constraints propagate with the fields: within those declared models, future sources do not alter past responses.

The work also extracts a reusable mathematical lesson. A complicated gauge theory can sometimes be embedded in a larger system whose causal behavior is easier to control. If the reduction back to the desired variables is local and preserves the constraints and physical comparison rule, the causal response can be transferred to the reduced theory. In plain language, one should judge causality using the complete network of fields, equations and redundancies—not necessarily one isolated fourth-order equation.

This is a classical result. It does not yet provide a complete interacting quantum theory, and it has not been proved for every curved spacetime.

A new changed-theory candidate has now passed the same kind of complete causal bookkeeping. It uses two internal phases moving in opposite directions and one auxiliary gauge connection. Their total diagonal gauge charge cancels, while their relative phase still changes and can act as a clock. On one selected squashed-sphere background, the complete gravity–clock–gauge system has an exactly checked 70-part causal gauge complex. In ordinary language: the equations, constraints and redundancies propagate together without using an instantaneous nonlocal shortcut.

This is the first proposed repair after the compensator failures to reach a complete classical causal parent. It is not yet a finished healthy universe. The exact stationary equations make the passing fixed-action Berger solution an isolated point rather than an open family, and the same action does not connect it continuously to the round cylinder. Its homogeneous trace motion has a healthy oscillatory sign, but every surviving nonhomogeneous physical direction still had to pass the full reduction. That reduction has now found a precise conflict. If the relative charge is fixed and its corresponding shift is treated as redundant, time translation becomes gauge-like—but the relative clock disappears with it. If the clock and its conjugate charge are kept, the clock survives, but time translation carries a physical charge.

This does not invalidate the causal equations. It says that this model cannot provide both a physical evolving clock and gauge time on the same declared phase space. The next calculation asks whether the unrestricted, charged-time version is otherwise healthy. That alternative is closer to ordinary physics, where time translations generate energy, but it is a different answer to the project's original question.

The first part of that health test is now exact. The clock's energy bends in the positive direction, and there is no exponentially growing solution in its homogeneous block. A small change in the clock's conserved momentum changes its rate, so its phase gradually drifts away from the original clock. The exact equations now show that this is the ordinary action–angle behavior of healthy clocks with slightly different rates, not a negative-energy mode. After allowing for the changed rate, the motion is orbitally stable. There is an important qualification: the complete gravity–clock background is an isolated solution, so the nearby family of clock rates exists in the reduced clock system rather than as a certified family of complete universes. Every nonhomogeneous wave sector must still pass its own health test. The first attempt to run that larger test found that the obvious dictionary is actually wrong rather than merely absent. Squashing the sphere mixes wave families that separate cleanly on a round sphere, so the simplest scalar/vector/tensor sorting is not preserved by the full gauge equations. This is not a bad wave or a failure of causality. The project is now building larger mixed blocks that the equations genuinely preserve before judging their health.

The original Berger laboratory adds something more tangible: a matter field acts as a clock, and an electromagnetic signal is emitted during a bounded time interval. Its response is retarded, and the frequency measured relative to the clock is unaffected by arbitrary coordinate, scale and electromagnetic gauge choices. One exact fixture gives a redshift ratio of (1+z=2).

That is a real relational redshift calculation in a model, but not yet a prediction for a star or galaxy. The attempt to promote its expanded apparatus has now produced an exact no-go inside the complete declared fixed-background, single-action local repair family: the necessary first-order gauge operation cannot be made to square to zero there. This does not undo the certified linear redshift experiment. It rules out that particular nonlinear apparatus completion and redirects the observer programme to the new two-phase counterflow parent, where the relative clock and detector must be constructed afresh from the changed action.

The changed clock now also has a finite-resolution theorem. A real clock cannot select an infinitely sharp instant, so the construction replaces the ideal slice by a narrow sampling window. The observable remains causal and its error decreases quadratically with the window width, provided a physical receiver survives the pending wave reduction. That condition has not yet been met, so this is a tested measurement rule rather than a new redshift result.

The comparison rules also work for chains of clocks. Two successive frequency ratios multiply to the direct ratio only when both comparisons use the same middle clock record and the same physical charge sector. Around a loop, any remaining factor records how the clock conventions or charge sectors were matched. If no such matching exists, the comparison is undefined rather than a mysterious physical effect. This is an exact consistency test for a future clock network, still conditional on finding a nonzero physical receiver in the changed theory.

2. Nonlinear balance removes some waves—and resonance adds another test

Linear equations describe very small disturbances. Nature is nonlinear: waves gravitate, light carries energy, and different fields interact. A linear solution is physically meaningful only if it can be continued into the nonlinear equations.

In the compact circle-by-sphere universe, the project has catalogued the certified finite families of ordinary Einstein-like waves, additional fourth-order waves and several exceptional global motions. At second order, five global balance conditions decide whether a finite combination has a formal continuation when controlled polynomial-in-time corrections are allowed.

This resembles balancing a spinning object. An individual motion may carry a nonzero global imbalance and fail, while a carefully chosen combination can cancel it. Some extra waves therefore disappear not because they were gauge, but because they cannot be the first step of an exact solution in that fixed charge sector.

Formal continuation is not the same as stable repeating motion. A correction may grow like time multiplied by an oscillation. The equations can still be solved, but the original small-wave approximation eventually breaks down. The project has found additional resonance conditions—forcing at a natural wave frequency—that are independent of the five global balances. Thus the emerging rule is:

> Global balance decides whether a formal next step exists; resonance helps decide whether that next step can remain bounded.

This is no longer based on one angular pattern. For the ordinary Einstein-like wave families, the same tuned balance-and-resonance geometry has now been proved for every angular momentum tested by the general formula. The first experiment combining two different absolute momenta has also computed all 164 planned basic couplings. Every basic coupling is individually obstructed, but combinations may still cancel; the full cancellation geometry remains open.

The interaction story has also produced a useful correction. An apparent gravity–light interaction survived an initial reduction. Once the missing gauge structure was included, however, that term could be removed through the first derivative order. A larger second-derivative calculation is now testing whether any invariant interaction remains. This is precisely why the project treats intermediate symbolic results as provisional until the full gauge structure has been checked.

These findings do not yet prove long-term stability. They turn a vague fear about higher derivatives into explicit balance and resonance tests that can pass or fail mode by mode.

There is also a deeper lesson about combining Einstein-like gravity with the larger Weyl theory. The linear embedding is exact, but at the first nonlinear order it develops a nonzero mismatch that cannot be repaired by a purely local correction on the complete compact wave space. Globally, that mismatch is measured by five conserved balances. Locally, however, the theory must first carry a flow of conserved current; the five charges appear only after that current is integrated over all of space. The canonical local current and its basic conservation identity have now been constructed from the full linear equations. The next test is to insert all five symmetries and prove that the integrated currents reproduce the five global balances.

There is now a second warning at the linear physical level. The simplest map identifies the Einstein-like solutions correctly as solutions of the larger equations, but it does not preserve the theory's own action-derived rule for pairing waves. In ordinary language, importing Einstein's equations is not yet the same as importing Einstein gravity as a complete physical subsystem. Before the remaining global charge reduction, the familiar and additional solution families nevertheless form an exact accounting: no linear solution is lost or double-counted. The project is now testing whether a corrected physical identification exists or whether the pairing mismatch is an invariant part of the larger theory.

3. Black-hole regularity does not trivially erase the additional family

The black-hole programme begins with an exact family of spherical solutions of pure Weyl gravity. It includes Einstein-like members and genuinely additional fourth-order geometries. In the declared static setting, the project obtains regular horizons, a consistent energy, horizon entropy and an exact first-law relation.

The more difficult question concerns waves around the Schwarzschild member. The familiar Einstein axial and polar gravitational waves are recovered, and the additional fourth-order content can be described through curvature waves. Those additional families have regular ingoing solutions at the future horizon.

The action-derived classical pairing gives an unusual result. The ordinary Einstein wave is self-silent in the pure-Weyl pairing, while controlled real-frequency calculations find nonzero pairing between the Einstein and additional waves and for a specified reconstruction of the additional metric wave. This shows that the additional solution is not merely a duplicated factor in the equation. The sign is a classical structural fact, not yet a positive quantum probability or energy statement.

The outer-boundary calculation has now gone one important step further for both axial and polar wave patterns. Reconstructing the metric from the additional curvature waves produces enhanced or logarithmic tails. In the tested case, those tails make their total symplectic size at infinity diverge, while the Einstein waves remain finite. Requiring finite size therefore selects exactly the Einstein family in both parities at that fixture—even though the additional families remain regular at the horizon.

The supported conclusion is therefore:

> The additional axial and polar families reach the horizon regularly, but a finite-size condition at infinity excludes them in one exactly checked fixture. Whether this Einstein selection holds for all frequencies and angular patterns remains open.

There is also a perfectly local linear initial-data restriction that sets the extra curvature and its initial velocity to zero. The unanswered questions are whether finite norm selects the same space generally, whether that space has the right nondegenerate wave structure, and whether interactions preserve it. General frequencies and angular patterns, complex frequencies, ringdown, stability and Hawking radiation remain future work.

The quantum fork

Quantum physics may decide the ghost question, but the project is not yet at a final quantum verdict.

The strict pure-Weyl theory develops a nonzero local anomaly at one loop. An anomaly means that a symmetry of the classical equations fails after the quantum calculation is regulated. Adding a conformal compensator can repair the tested local quantum identity at that order—but adding a compensator changes the theory. The honest conclusion is not "the anomaly is solved," but that the strict and compensated theories occupy different branches of the programme.

Adding ordinary healthy conformal matter does not cancel the strict anomaly: one exact anomaly coordinate has the same positive sign for the gravitational field and for every standard scalar, fermion and gauge-vector contribution. Putting those fields into compact gauge-group representations cannot help, because forgetting the gauge labels maps them back into the same already empty nonnegative cone. This is an incompatibility result, not a selection of the Standard Model, a grand unified group or a particle spectrum; no representation scan is being advertised.

On the closed spherical background, a reduced calculation gives another clue. After gauge and constraint reduction, the candidate isolated conformal-graviton class disappears. Two collective curvature classes remain, but they have not been interpreted as particles.

A separate reduced free-wave construction produces candidate two-point functions with the required short-distance behavior. The Einstein-like family has a positive sign, while two additional families have negative signs. This is a concrete warning. It is not yet a particle theorem because the complete quantum gauge reduction, interacting observables and positive probability space have not been built.

Several outcomes remain possible. The negative directions might be removed by the final constraints, paired differently in a valid quantum construction, confined by nonlinear conditions—or survive as genuine ghosts and rule out the theory.

A compact public scorecard

The programme is deliberately incomplete. The table below separates results that exist in a declared model from the physical conclusions still being tested.

Demonstrated in a model	Partly demonstrated	Open
Complete classical causal propagation on several controlled backgrounds, including a changed two-phase counterflow parent	A causal clock-defined redshift signal; the changed model has a charged homogeneous clock and a finite-resolution measurement rule, but its obvious round-sphere wave sorting fails on the squashed background and must be replaced	Electrons, mass generation and Standard Model matter
Gauge reduction and exact classical comparison rules	Additional compact and black-hole wave families, with their final physical status unresolved	A positive interacting quantum state and physical particles
Global nonlinear balance conditions and explicit resonance obstructions	Static black-hole thermodynamics; additional waves reach the horizon but are excluded by finite size at infinity in both parities at one fixture	Full black-hole scattering, stability, ringdown and Hawking radiation
A one-loop anomaly calculation and a compensator-based local repair in a changed theory	Candidate reduced quantum two-point functions with both positive and negative signs	Gravitational lensing, realistic cosmology, dark matter and dark energy

Light, gravitational waves, redshift and black holes therefore occur in precisely scoped mathematical forms. Photons, gravitons and electrons do not yet occur as established particles. Gravitational lensing, galaxy dynamics and cosmology are later observational tests, not current achievements.

The live universe-building map and the searchable residual atlas carry the detailed status and update as new certificates land.

How AI-assisted research is being tested

This is also an experiment in scientific method. The programme is directed by Asger Alstrup Palm, Honorary Professor at DTU Compute, whose field is computer science rather than physics. AI agents perform much of the detailed symbolic derivation, adversarial review, literature mapping, verification and drafting. Palm chooses the overall direction and retains editorial responsibility for the public claims. His DTU affiliation identifies his academic role; it does not imply that DTU endorses the project or its conclusions.

The calculations are too large and too sensitive to signs and conventions to manage reliably as hand algebra alone. Some contain hundreds of linked field and constraint components and more than one hundred thousand symbolic interaction terms. Size is not evidence. The important point is that the terms are generated from declared equations and then subjected to independent checks.

The repository functions like a symbolic experimental laboratory. A material claim is expected to carry:

1. its theory, background, charge sector, boundary conditions and claim level;
2. a machine-readable certificate recording inputs, assumptions, hashes, outputs and unresolved fields;
3. an independent verifier that checks the decisive identity rather than trusting the program that generated it;
4. deliberately damaged inputs that must fail, showing that the verifier is actually testing something;
5. a record of convention errors, negative results and withdrawn interpretations rather than a cleaned-up success story;
6. a clean, reproducible route from the archived calculation to the public paper.

This discipline has already mattered. Cyclic sign conventions in the interaction calculation required repair. A black-hole causal claim was narrowed when a repeated asymptotic root was examined more carefully. An apparent invariant gravity–light interaction became removable after the missing gauge direction was restored. These are not embarrassments to hide; they are examples of the accountability system doing its job.

Code cannot prove assumptions that were never encoded. Global existence, physical boundary conditions, experimental relevance and quantum interpretation must each pass their own tests. The system is designed to stop a local algebraic success from being silently advertised as a result about particles or nature.

What would count as success—or failure?

The programme does not need Weyl gravity to win.

One possible result is **Einstein recovery**: the extra solutions are removed by physically justified constraints, boundaries or quantum reduction, while ordinary relativity survives with the correct observables.

Another is **healthy new physics**: an additional wave, interaction or cosmological effect survives causality, nonlinear stability, observation and quantization. Only then would it become a credible prediction.

A third is **sector-dependent viability**: the theory works on a closed, zero-charge universe but not when time translation carries energy at an outer boundary, or it works only with an explicitly stated compensator or boundary condition. That is not a contradiction. It tells us which theory and which physical sector are actually under discussion.

The fourth outcome is an **exact no-go**: an unavoidable anomaly, negative physical state, instability or boundary obstruction ends the candidate theory under a complete set of assumptions.

The project can stop at any rung and still produce useful physics. A precise failure tells us which attractive idea cannot describe nature, under which conditions, and what a successful replacement would have to change. A passed rung narrows the alternatives and makes the next test meaningful.

The central question remains open:

> Are Weyl gravity's additional solutions redundant descriptions, constrained waves, boundary excitations, healthy new physics—or genuine ghosts?

We are building model universes carefully enough that each of those answers can be tested rather than assumed.

For readers who want the technical continuation

This article is the public front door. The physicist executive summary provides the claim-by-claim result spine, exact scope labels and links to papers and certificates. A PDF version is available as a short technical briefing.

The public construction map shows dependencies between familiar physical milestones. The residual atlas provides the detailed, searchable ledger of wave families and their causal, symplectic, nonlinear, observational and quantum status.